Rasheed Abdul Haq Kozhiparamban

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Summary

- Senior AI/ML Engineer with a **Ph.D.** in Artificial Intelligence and 8+ years of experience across AI research and applied machine learning, spanning **Computer Vision**, **Industrial AI**, **Generative AI**, **Predictive Analytics**, and **Time-Series Forecasting**.
- Proven expertise in designing and deploying end-to-end AI solutions spanning data engineering, feature engineering, model development, validation, MLOps, inference, and human-in-the-loop workflows for enterprise and industrial applications.
- Hands-on experience with **Python**, **PyTorch**, **TensorFlow**, **OpenCV**, **YOLOv8/YOLO11**, **Azure ML**, **AWS Bedrock**, and modern LLM ecosystems for scalable AI development and deployment.
- Author of peer-reviewed IEEE publications and three granted Indian patents in Deep Learning, Virtual Sensing, and Predictive Analytics.

Education

- **Ph.D.** in Artificial Intelligence (Deep Learning, Predictive Modeling & Time-Series Forecasting), National Institute of Technology (NIT), Puducherry, India, 2018 – 2024
- **M.Tech** in Electronics Design Technology, National Institute of Electronics & Information Technology (NIELIT), Calicut, India, 2012 – 2014
- **B.Tech** in Electronics and Communication Engineering, School of Engineering, CUSAT, Cochin, India, 2005 – 2009

Experience

Chevron Engine, Bengaluru, India: Senior Machine Learning Engineer
July 2025 – Present

- Developed production-grade **computer vision** solutions for industrial inspection, asset integrity, and worker safety in oil & gas, spanning data ingestion, annotation, model training, validation, deployment, and human-in-the-loop review.
- Led the AI/ML handover of a production drone-inspection platform to an in-house team, defining data-readiness and knowledge-transfer criteria and supporting end-to-end **user acceptance testing (UAT)** through sign-off. Contributed to serverless architecture simplification and assessed user-triggered inference workflows.
- Designed a two-stage component-segmentation and corrosion-detection pipeline to suppress background false positives; aligned independently versioned **Azure ML** endpoints and inference orchestration. Supported model training, registry, data-platform integration, and production monitoring.
- Built high-resolution inspection workflows with overlapping image tiles, prediction stitching, configurable class thresholds, and structured outputs, displaying merged detections as continuous annotations for inspector review.
- Improved utility-asset component segmentation through dataset audits, class-imbalance analysis, annotation-quality reviews, class refinement, and preprocessing recommendations; established labeling standards, validation gates, and subject matter expert (SME) review workflows.
- Delivered worker detection and developed safety-focused video analytics addressing activity recognition, personal protective equipment (PPE) visibility, camera placement, occlusion, and data quality; packaged a **YOLO11 fire-and-smoke detection model** for edge-device testing.
- Advanced **digital twin** initiatives through sensor-data integration with virtual asset models for condition monitoring, work planning, and troubleshooting; supplied annotated inspection outputs for a 3D integration pilot.
- Built a physics-based **thermal tank-level estimation prototype** using radiometric calibration, robust temperature profiles, and gradient detection; delivered a web demo with weak-signal handling. Ground-truth validation remains pending.
- Advised on a **change-detection proof of concept** using image alignment, visual similarity, and vision-language models; recommended evaluation of classical alternatives and production cost and licensing checks.

- Evaluated and shortlisted computer vision vendors for environmental, health, and safety (EHS) use cases, assessing model accuracy, latency, hazardous-area certifications (ATEX/IECEx), edge deployment, data privacy, and workflow integration.
- Authored an annotation handbook, architecture references, and training material; co-authored enterprise CV deployment guidelines covering risk-based metrics, verification, human accountability, change control, and lifecycle governance.
- Led ML knowledge transfer, sprint planning, roadmap proposals, stakeholder demonstrations, and incremental SME model reviews, collaborating with product, software, data engineering, operations, inspectors, and vendors on deployment and future CV opportunities. Conducted Senior ML Engineer interviews using structured technical questions and evaluation rubrics.

SONY, Bengaluru, India: Lead AI/ML Engineer
April 2025 – June 2025 (3-month contract engagement)

- Built secure **RAG pipelines** using AWS Bedrock and Claude with enterprise data from **SharePoint and Confluence** via MCP, enforcing citation tracking and zero internet leakage.
- Integrated and tuned **Jira/Confluence MCP clients**, using AWS Secrets Manager for tokenized access and validated `--space/project` parameters.
- Developed **LLM tools for task analysis** and automated risk scanners, streamlining deployments across Windows and AWS EC2 environments.

DIMAAG-AI, Bengaluru, Karnataka, India: Senior Software Engineer AI/ML
February 2023 – February 2025 (2 years 1 month)

- Developed production-ready computer vision solutions for industrial and agricultural applications, leveraging **YOLOv8** for object detection and vision-LLM pipelines for advanced feature extraction.
- Deployed deep learning techniques for real-time **action recognition** and video analytics, implementing models such as **SlowFast**.
- Created methods for **Field of View (FOV)** validation to maintain camera alignment, combining feature matching, optical flow, and statistical comparisons.
- Built AI pipelines for **object segmentation**, classification, and feature extraction using segmentation models (**Unet**, **DeepLabV3**, **PSPNet**).
- Processed **3D point clouds** to generate segmentation masks, delivering accurate feature extraction for automation systems.
- Applied **Random Forest**, **Gradient Boosting**, and **MLP** classifiers for robust, high-accuracy predictions across diverse datasets.
- Enhanced data quality and balance via **image clustering** and model evaluation techniques, followed by **data augmentation** and hyperparameter tuning to improve model performance.
- Integrated **statistical analysis** and geometric transformations to optimize algorithmic precision for production environments.
- Implemented **LLM-based retrieval systems** to extract structured information from PDFs using **Vector Databases** (FAISS, Pinecone) and **Retrieval-Augmented Generation (RAG)**.
- Orchestrated **agentic workflows** via **LangChain** for dynamic query generation, structured output validation, and multi-step reasoning with LLMs.
- Developed prototypes leveraging **Generative AI** for summarizing SOP documents and automating question-answering systems, ensuring secure role-based access to sensitive data.
- Utilized **Azure AI services** (Azure Machine Learning, Cognitive Services) to streamline model training, deployment, and monitoring.
- Deployed **MLOps** workflows using **MLflow** for experiment tracking, model versioning, and containerized deployment on **Azure**.
- Drove end-to-end design, execution, and evaluation of AI model experiments, collaborating with cross-functional teams and mentoring junior engineers on deep learning best practices.

National Institute of Technology, Puducherry, India: Senior Research Fellow
July 2018 – January 2023 (4 years 7 months)

- Developed virtual sensing and predictive analytics solutions for multivariate time-series forecasting of water quality parameters using deep learning.
- Engineered and optimized deep learning models (LSTM, GRU, Bi-LSTM) for multivariate time-series forecasting, improving prediction accuracy for aquaculture management.
- Proposed and implemented hybrid CNN-LSTM and CNN-GRU architectures for real-time predictive modeling, balancing forecasting accuracy and computational efficiency.
- Researched hyperparameter optimization (learning rate, epochs, batch size) to maximize model performance, achieving significant reductions in computation time.
- Applied **attention mechanisms** to deep learning models to enhance the accuracy of water quality predictions by focusing on the most relevant features in time-series data.
- Published peer-reviewed journal papers on predictive modeling, virtual sensing, and deep learning for time-series forecasting.
- Filed and was granted **patents** for new methods in virtual sensing and hybrid deep learning systems for aquaculture water quality monitoring.
- Conducted real-world experiments on aquaculture datasets (e.g., ADAK and MAC datasets) to validate the accuracy of the developed models.
- Designed and implemented an IoT-based low-cost water quality monitoring system, using sensors and cloud-based virtual sensing to provide real-time data for farmers.
- Built and validated water quality estimation models based on low-cost sensor data, reducing dependency on expensive sensors while maintaining high accuracy in dissolved oxygen prediction.

Patents:

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF", Indian Patent No.: 397595, Date of Grant: May 25, 2022.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING", Indian Patent No.: 552671, Date of Grant: October 21, 2024.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P. , "HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION", Indian Patent No.: 579506, Date of Grant: January 30, 2026.

Publications:

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. "Deep Learning Enhanced Virtual Sensing: A Cost-Effective Approach for Water Quality Monitoring in Aquaculture" *IEEE Transactions on AgriFood Electronics*, June 2025.
2. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models", *IEEE Access*, vol. 10, pp. 60078-60098, June 2022.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Water Quality Prediction System based on Adam Optimised LSTM Neural Network for Aquaculture: A Case Study in Kerala, India", *J. Inst. Eng. India Ser. B*, *Springer*, 103, 2177–2188, October 2022.
4. Rasheed Abdul Haq K. P. and Harigovindan V. P., "Accurate Dissolved Oxygen Prediction for Aquaculture Using Stacked Ensemble Machine Learning Model", *Natl. Acad. Sci. Lett.*, *Springer*, 46, 203–207, January 2023.
5. Rahul Gandh D, Harigovindan V P, Rasheed Abdul Haq K P, and Amrtha Bhide, "Attention-driven LSTM and GRU deep learning techniques for precise water quality prediction in smart aquaculture", *Aquaculture International*, vol. 32, pp. 8455–8478, 2024. doi:10.1007/s10499-024-01574-5.

MEA Engineering College, Perinthalmanna, Kerala, India: Assistant Professor, Department of ECE
June 2015 – July 2018 (3 years 1 month)

- Delivered undergraduate and postgraduate courses in Digital Signal Processing, Signals & Systems, Digital System Design, Basics of Electronics Engineering, Electronic Circuits, and Communication Systems, while developing structured course materials aligned with the APJ Abdul Kalam Technological University (KTU) curriculum.

- Led the department's transition from the Calicut University syllabus to the KTU curriculum by preparing course files, lecture notes, assessment plans, question banks, laboratory manuals, and outcome-based evaluation documents.
- Served as Class Tutor for the 2014–2018 Electronics & Communication Engineering batch, managing student mentoring, attendance analytics, academic performance tracking, parent interactions, and intervention plans for academically at-risk students.
- Coordinated B.Tech Mini Projects, B.Tech Seminars, and M.Tech Seminar activities by organizing student groups, assigning faculty mentors, scheduling evaluations, and consolidating academic assessments.
- Guided undergraduate engineering projects in embedded systems and wireless technologies while mentoring students in technical writing, project execution, and research presentation.
- Conducted laboratory sessions in Basic Electronics, Linear Integrated Circuits, and Communication Systems, and served as an External Laboratory Examiner for Electronics laboratories under the university examination system.
- Developed spreadsheet-based academic tracking systems for attendance, internal assessment, and departmental reporting, improving the efficiency and accuracy of academic administration.

Way in Electronic Solutions, Calicut, India: Founder & Senior Embedded Software Engineer
August 2014 – June 2017 (Part-time while serving as Assistant Professor) (2 years 11 months)

- Co-founded an embedded systems startup delivering custom industrial automation, smart parking, medical-device testing, and intelligent monitoring solutions from concept through deployment.
- Led end-to-end product development, including requirements analysis, system architecture, embedded firmware development, hardware integration, testing, technical documentation, deployment, and customer support.
- Designed and developed embedded products including the Intelligent Parking Guidance System (IPGS), Ultrasound Probe Test System, GSM-based Security System, Digital Signage solutions, and Animal Health Monitoring System (AHMS).
- Engineered Embedded C firmware for 8051-based systems and Raspberry Pi applications, integrating RS-485/UART communication, ultrasonic sensors, LED driver interfaces, and Linux-based embedded platforms.
- Built web-based monitoring and management applications using PHP, MySQL, JavaScript, HTML/CSS, and XAMPP for real-time visualization, remote monitoring, and data management.
- Authored high-level design documents (HLDs), coding standards, technical proposals, quotations, business plans, and customer presentations while supporting product planning and engineering best practices.
- Collaborated directly with customers, technology partners, and multidisciplinary teams to translate business requirements into deployable engineering solutions while contributing to startup operations, recruitment, and business development.

M. N. Dastur & Co. (P) Ltd., Bengaluru, India: DSP Intern (Embedded Image Processing)
September 2013 – July 2014 (11 months)

- Engineered an Automatic Number Plate Recognition (ANPR) system for Indian vehicles using Computer Vision and Digital Image Processing as part of an M.Tech internship project targeting embedded DSP/ARM platforms.
- Designed and implemented the complete recognition pipeline covering license plate localization, character segmentation, Optical Character Recognition (OCR), and Support Vector Machine (SVM)-based classification using MATLAB, OpenCV, and C++.
- Implemented image preprocessing, Sobel edge detection, Otsu thresholding, morphological operations, contour analysis, histogram equalization, affine rotation correction, and feature extraction to improve recognition robustness.
- Ported and evaluated the OpenCV implementation on Ubuntu Linux running on the ARM-based BeagleBoard-xM platform, addressing embedded deployment and cross-compilation challenges.
- Authored core C++ modules, MATLAB algorithms, training utilities, technical documentation, and project reports while validating the solution through iterative prototype development and engineering reviews.

Bharti Airtel Ltd., Kerala, India: Network Surveillance Engineer
January 2010 – February 2012 (2 years 2 months)

- Monitored nationwide telecom network health, service availability, alarms, and key performance indicators (KPIs) using Operations & Maintenance Center (OMC-R) network management tools.
- Performed network fault detection, troubleshooting, root cause analysis, and performance optimization by coordinating with field engineers, vendors, and cross-functional operations teams to minimize service disruptions.

- Coordinated planned network outages, preventive maintenance activities, and vendor-recommended corrective actions while ensuring controlled execution with minimal customer impact.
- Provided remote technical support for fault investigation, managed alarm escalation processes, classified network outages, and ensured timely incident resolution in accordance with established escalation procedures.
- Analyzed network alarm trends and generated daily, weekly, and monthly operational reports to support performance monitoring, reliability improvements, and continuous network optimization.

Personal AI Projects

Arogyamitra — AI medical triage assistant for Telegram (Kerala, India): **2025**
LLM-driven chatbot that collects demographics and symptoms, triages risk, and recommends specialists.

- Delivered production-ready Telegram bot with structured multi-field demographic and symptom collection.
- Implemented multi-provider LLM fallback and persistent rate-limit tracking for robust uptime.
- Built precise doctor-matching with exclude-keyword logic to prevent misclassification.

Stack: Python, python-telegram-bot, Groq (LLM), OpenRouter/Gemini, HuggingFace, CSV (data), SQLite (planned), Railway (deployment) Links: GitHub | Demo

Skills

- **Programming Languages:** Python, C++, MATLAB
- **AI/ML Expertise:** Predictive Modeling, Time-Series Forecasting, Statistical Learning, Machine Learning, Deep Learning, Computer Vision, Thermal Image Analysis, Instance Segmentation, Change Detection, Vision-Language Models, Digital Twins, NLP, Generative AI, LLMs, Retrieval-Augmented Generation (RAG), Agentic AI
- **Libraries/Frameworks:** PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, YOLOv8/YOLO11, PyTorchVideo, Hugging Face Transformers, LangChain, NumPy, Pandas, Matplotlib, NLTK, spaCy
- **MLOps & Cloud:** Azure ML, Managed Endpoints, Model Registry, Multi-model Orchestration, Azure Functions, Azure AI Services, AWS Bedrock, MLflow, Git
- **Vector Databases & Retrieval:** FAISS, Pinecone
- **Operating Systems:** Linux, Windows
- **Professional Skills:** Technical Leadership, Cross-functional Collaboration, Technical Mentoring, Stakeholder Communication, Annotation Standards, Model Verification, UAT, Technical Documentation

References

Available upon request.